Absolutely maximally entangled states with local dimensions 8 and 16, a coupling rule for composite dimensions, and a confirmation that no qubit-pair stabilizer AME(8,4) state exists

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An n-party pure state of local dimension d is absolutely maximally entangled, AME(n, d), if every set of $\lfloor n/2 \rfloor$ parties is maximally mixed. Which pairs (n, d) admit such states is open in general. We construct AME$(15, 8)$ and AME$(17, 8)$, two cells left blank (existence unknown) in the Huber–Wyderka table of AME states (version of 8 June 2026), as graph states whose adjacency matrix is a symmetric circulant over $\mathbb{F}_8$. An exhaustive scan of all symmetric circulants over $\mathbb{F}_8$ finds 336 and 48 solutions up to scaling, in 28 and 4 orbits under multipliers and field automorphisms. Beyond that table, which stops at $d = 11$, we construct AME$(n, 16)$ for $n = 18, 19, 20, 21$, cells not given by the published constructions in our archived snapshot, from circulant graph states over $\mathbb{F}_{16}$ and from Hermitian self-dual MDS codes over $\mathbb{F}_{256}$ that are not generalized Reed–Solomon codes. We also state a coupling rule: a k-uniform state of $2k + 2$ parties of local dimension a and an AME$(2k + 2, b)$ state with $a \leq b$ give an AME$(2k + 1, ab)$ state. With explicit ingredients it yields 26 cells AME$(11, d)$, AME$(13, d)$ and AME$(15, d)$ with composite $d \leq 62$ that the published constructions in the snapshot do not give. All 32 states were certified by a frozen exact verifier (for the coupling cells through exact checks of both ingredients) and re-checked with independent code. For AME$(8, 4)$, the smallest open case, we show that an AME$(8, 4)$ stabilizer state with respect to a group-type nice error basis would have to be an $\mathbb{F}_2$-linear (qubit-pair) stabilizer state, and we confirm the theorem of Ball, Moreno and Simoens that no such state exists by a differently organized exhaustive computer search, implemented twice with independent code. These statements concern stabilizer states in the group-type nice-error-basis model only; AME$(8, 4)$ remains open, for non-stabilizer states and for error bases outside that model. The results were obtained by a multi-agent language-model system working against a frozen exact verifier, and all certificates and ingredient matrices, the exhaustive searches behind every count and nonexistence statement, and independent checkers are released with a replay script.

I. INTRODUCTION

A pure state $|\psi\rangle \in (\mathbb{C}^d)^{\otimes n}$ is *absolutely maximally entangled*, AME(n, d), if $\rho_A = \mathbb{1}/d^{|A|}$ for every set A of $|A| = \lfloor n/2 \rfloor$ parties. Equivalently it is a perfect tensor, or a pure quantum code $((n, 1, \lfloor n/2 \rfloor + 1))_d$. AME states are the resources of quantum secret sharing and parallel teleportation [1], and the building blocks of holographic codes [2]. They are closely tied to orthogonal arrays and MDS codes [3, 4]. Their existence is a long-standing open problem (problem 35 of the IQOQI list; see the review [5]). Nonexistence results come from the shadow inequalities [6, 7], Scott's bound [8], and special arguments for four and seven qubits [9, 10]. Constructions come mostly from stabilizer codes and quantum MDS codes [4, 11–13], and from a few non-stabilizer states such as the "golden" AME$(4, 6)$ [14].

The Huber–Wyderka table [15] records the status of every cell with $d \leq 11$. Three 2026 preprints settled further cells: all seven-party cases $d \geq 3$ [16], AME$(12, 5)$, AME$(18, 11)$ and AME$(18, 13)$ via Hermitian self-dual MDS codes [17], and AME$(2q, q)$ for every odd prime power q [18]. Kitamura settled two more cells, with machine-checked proofs in Lean (public repositories, not refereed): a non-stabilizer AME$(9, 6)$ [19], also discussed in Ref. [20], and AME$(11, 10)$ [21], which couples a 12-qubit state with an AME$(12, 5)$ state (Sec. V). The kernel of Ref. [18] has no direct analogue for $q = 2^r \geq 8$ (their Lemma S4), and the even prime powers $d = 4, 8$ remain the least understood. The first open case overall is AME$(8, 4)$ [5, 16]. For additive (qubit-pair) stabilizer states it was settled in the negative by Ball, Moreno and Simoens [22], who show that no $[[8, 0, 5]]_4$ stabilizer code exists.

This paper reports three groups of new states and one independent confirmation. Section III constructs AME$(15, 8)$ and AME$(17, 8)$. Both cells are blank in the Huber–Wyderka table (Table I); the code tables list $[[15, 0, d]]_8$ with $7 \leq d \leq 8$ and $[[17, 0, d]]_8$ with $8 \leq d \leq 9$, while AME needs $d = 8$ and $d = 9$ [23]. The states are qudit graph states with circulant symmetry over $\mathbb{F}_8$, and a simple counting heuristic suggests why these two cells, and no neighbouring ones, fall to this method. Section IV goes beyond the table, which stops at $d = 11$: it constructs AME$(n, 16)$ for $18 \leq n \leq 21$, where Reed–Solomon codes stop at $n = 17$. Section V states a coupling rule that glues a k-uniform state of dimension a to an AME state of dimension b and returns an AME state of dimension ab; Kitamura's AME$(11, 10)$ is one instance, and with explicit ingredients the rule gives 26 further cells with composite d. Section VI turns to AME$(8, 4)$. We reduce every group-type stabilizer formalism for ququarts to qubit pairs (Proposition 1), and we confirm the theorem of Ref. [22] for qubit pairs by an

TABLE I. Existence of AME(n,d) for $13 \le n \le 22$ and $d \in \{4,5,7,8,9,11,16\}$, after this work ($d = 2$ and 3 are omitted; there AME$(15,3)$ is the only cell in this range not covered by the snapshot). The row $d = 16^\dagger$ lies beyond the Huber–Wyderka table; there ✓ means that a published family gives the state (Sec. IV). Sources: the Huber–Wyderka table [15] (updated 8 June 2026) and the 2026 preprints [16–18] (unrefereed). ✓ known to exist, ★ this work, × known not to exist, ? not covered by the snapshot (archived with the release; a statement about the snapshot, not about the whole literature). Ref. [20] announces (without a construction; an announcement, not a published theorem) that a forthcoming revision will include stabilizer AME(N,q) for all $2 \le N \le 2q$, without stating the range of q (Ref. [18] treats odd q). For $q = 11$ this would cover AME$(19,11)$ and AME$(20,11)$; for $q = 8$ it would cover AME$(15,8)$ and AME$(16,8)$, and for $q = 16$ every $N \le 32$, including the four cells of Sec. IV and AME$(22,16)$. Over all cells with $2 \le d \le 11$ up to Scott's bound, the snapshot (including Kitamura's two cells and our AME$(15,8)$ and AME$(17,8)$) has 118 existing and 96 nonexistent cells, and 906 cells that it does not cover.

$d\backslash n$	13	14	15	16	17	18	19	20	21	22
4	?	?	?	×	?	?	?	×	?	?
5	✓	✓	✓	?	✓	✓	?	?	?	?
7	✓	✓	✓	?	✓	✓	?	?	?	?
8	✓	✓	★	?	★	?	?	?	?	?
9	✓	✓	✓	✓	✓	✓	?	?	?	?
11	✓	✓	✓	✓	✓	✓	?	?	✓	✓
$16^\dagger$	✓	✓	✓	✓	✓	★	★	★	★	?

exhaustive search that two independent programs carry out (Sec. VI B). Together these exclude every group-type stabilizer AME$(8,4)$ state. Appendix B gives an earlier analysis of the symmetric sectors, which now serves as a cross-check. Section VII describes how the results were obtained and checked.

II. STABILIZER AME STATES AND GRAPH STATES

Let $d = q = p^m$. A qudit stabilizer state on n parties is described by an $\mathbb{F}_p$-linear subspace $S \subset \mathbb{F}_p^{2mn}$ of dimension mn. Each party contributes m coordinates for X and m for Z, and S is isotropic for the symplectic form $\omega((a|b),(a'|b')) = a\cdot b' - b\cdot a'$ [24]. For a set A of parties, $\rho_A = d^{-|A|}\sum_{g\in S,\ \mathrm{supp}\, g\subseteq A} \chi(g)\, g|_A$ for a character χ, so ρ_A is maximally mixed exactly when no nonzero element of S is supported inside A. Writing S as the row space of an $mn \times 2mn$ matrix M, the elements supported in A form a space of dimension $mn - \operatorname{rank} M[\,\cdot\,, A^c]$, where $M[\,\cdot\,, A^c]$ keeps the X and Z columns of the parties outside A. This gives the criterion used throughout:

Lemma 1. *The stabilizer state of S is AME(n,p^m) if and only if, for every set B of $\lceil n/2\rceil$ parties, the $mn \times 2m\lceil n/2\rceil$ column block $M[\,\cdot\,, B]$ has rank mn over $\mathbb{F}_p$.*

An $\mathbb{F}_q$-linear graph state is given by a symmetric matrix $\Gamma \in \mathbb{F}_q^{n\times n}$ with zero diagonal. For $q = 2^m$, write Tr for the trace $\mathbb{F}_q \to \mathbb{F}_2$. The state is

$$|\Gamma\rangle = q^{-n/2}\sum_{x\in\mathbb{F}_q^n} (-1)^{\mathrm{Tr}\left(\sum_{i<j}\Gamma_{ij}x_ix_j\right)}|x\rangle, \qquad (1)$$

stabilized by $X(\lambda e_i)Z(\lambda\Gamma_i)$ for $\lambda \in \mathbb{F}_q$, where $X(a)|x\rangle = |x+a\rangle$ and $Z(b)|x\rangle = (-1)^{\mathrm{Tr}(b\cdot x)}|x\rangle$ [25–27]. An element $\sum_i X(c_ie_i)Z(c_i\Gamma_i)$ is supported in A only if $c_i = 0$ for $i \notin A$ and $c_A\Gamma[A,A^c] = 0$. Hence:

Lemma 2. *$|\Gamma\rangle$ is AME(n,q) if and only if $\Gamma[A,A^c]$ has full row rank $\lfloor n/2\rfloor$ over $\mathbb{F}_q$ for every set A of $\lfloor n/2\rfloor$ parties.*

Expanding $\mathbb{F}_q$ in an $\mathbb{F}_p$-basis for the X part, and in its trace-dual basis for the Z part, turns the $\mathbb{F}_q$-symplectic structure into the $\mathbb{F}_p$-symplectic one of Lemma 1 [24]. This is the form our verifier checks. An $\mathbb{F}_{q^2}$-linear code $C \subset \mathbb{F}_{q^2}^{2k}$ that is Hermitian self-dual (self-dual for $\langle x,y\rangle = \sum_i x_iy_i^q$) gives in the same way an $\mathbb{F}_q$-linear stabilizer state of $2k$ parties of dimension q, and this state is AME$(2k,q)$ exactly when C is an MDS code $[2k,k,k+1]_{q^2}$ [11, 24]. With a generator matrix $[I\,|\,A]$ this means $A\bar{A}^T = -I$, with $\bar{A}$ the entrywise conjugate $x \mapsto x^q$, and that every square submatrix of A is nonsingular. Projecting one party of an AME$(2k,d)$ state onto a basis state gives an AME$(2k-1,d)$ state [5, 18]. The converse step, from odd to even n, does not hold in general.

III. AME(15,8) AND AME(17,8)

Write $\mathbb{F}_8 = \mathbb{F}_2[\alpha]/(\alpha^3+\alpha+1)$. A symmetric circulant on $\mathbb{Z}_n$ is $\Gamma_{ij} = c_{\delta(i,j)}$ with $\delta(i,j) = \min(|i-j|,\, n-|i-j|)$ and $c = (c_1,\ldots,c_{\lfloor n/2\rfloor})$.

Theorem 1. *The graph states (1) with symmetric circulant Γ over $\mathbb{F}_8$ and*

$$n = 15: \quad c = (1,\,0,\,\alpha^2,\,\alpha^4,\,\alpha^3,\,0,\,0),$$
$$n = 17: \quad c = (1,\,\alpha,\,\alpha,\,1,\,\alpha,\,\alpha^6,\,\alpha^6,\,\alpha)$$

are AME$(15,8)$ and AME$(17,8)$, respectively. Equivalently, they are pure stabilizer codes $[[15,0,8]]_8$ and $[[17,0,9]]_8$.

Verification. By Lemma 2, it suffices to check that all 6,435 matrices $\Gamma[A,A^c]$ (7×8) for $n = 15$, and all 24,310 matrices (8×9) for $n = 17$, have full row rank over $\mathbb{F}_8$. We did this with three programs that share no code, except that (iii) reuses the verifier's $\mathbb{F}_2$ row reduction to compare row spaces. (i) The frozen verifier of the search (Appendix A) checks the $3n$-qubit $\mathbb{F}_2$ stabilizer matrix stored in the candidate file: isotropy and rank in Python, all $\binom{n}{\lceil n/2\rceil}$ column blocks of Lemma 1 by Gaussian elimination in C, and a seeded sample of blocks again

in Python. (ii) An auditing agent wrote its own $\mathbb{F}_2$ bit-vector elimination over all blocks, together with an $\mathbb{F}_8$-level rank check of every $\Gamma[A, A^c]$. As a negative control for $n = 15$, exchanging one qubit between two parties produced 268 failing sets. (iii) The script accompanying this paper rebuilds Γ from the coefficients above with its own $\mathbb{F}_8$ arithmetic, checks every $\Gamma[A, A^c]$, and confirms that its $\mathbb{F}_2$ expansion spans the same stabilizer group as the certified candidate files. □

How they were found. The search scanned *all* symmetric circulants over $\mathbb{F}_8$ with first nonzero coefficient 1: 299,593 for $n = 15$ (336 AME states) and 2,396,745 for $n = 17$ (48 AME states). Each scan takes seconds in C. Every hit was re-verified. The maps $i \mapsto ui$ ($u \in \mathbb{Z}_n^\times$, a party permutation), $c \mapsto c^2$ (a local relabelling, since $\mathrm{Tr}(z^2) = \mathrm{Tr}(z)$), and $c \mapsto \lambda c$ (local scaling, since every $\lambda \in \mathbb{F}_8^\times$ is a square) preserve the AME property. On normalized vectors, scaling is absorbed and the group generated by multipliers modulo ± 1 and Frobenius has order 12 for $n = 15$ and 24 for $n = 17$. The solutions fall into 28 orbits for $n = 15$ and 4 for $n = 17$, all of size 12. For $n = 17$ this is half the group order: every solution, including the one in Theorem 1, is invariant under the multiplier $i \mapsto 4i \pmod{17}$, so its party-permutation symmetry contains $\mathbb{Z}_{17} \rtimes \mathbb{Z}_4$. Orbits may still be locally equivalent to one another, so these are upper bounds on the number of inequivalent states in the family.

Circulant matrices were used before by Ball [13] to construct Hermitian self-dual MDS codes over $\mathbb{F}_{q^2}$, that is, $\mathbb{F}_{q^2}$-linear AME states with an even number of parties; that search found $[[10, 0, 6]]_3$, $[[10, 0, 6]]_4$, $[[14, 0, 8]]_5$, $[[18, 0, 10]]_5$ and $[[18, 0, 10]]_7$ codes, among others, and no symmetric circulant for $q = 8$ and 18 parties. Our states are $\mathbb{F}_8$-linear graph states with an odd number of parties, and they cannot have been obtained by projecting states of that kind, since no AME$(16, 8)$ or AME$(18, 8)$ is known. The method, however, is closely related.

Why these two cells. For a uniformly random $\mathbb{F}_q$-linear $[[n, 0]]_q$ stabilizer state, i.e. a random Lagrangian subspace of $\mathbb{F}_q^{2n}$, a fixed nonzero vector lies in the subspace with probability $\prod_{i<n}(q^i + 1)/\prod_{i\le n}(q^i + 1) = 1/(q^n + 1)$, so the expected number of nonzero stabilizers of party weight at most $\lfloor n/2 \rfloor$ is $E = \sum_{w=1}^{\lfloor n/2 \rfloor} \binom{n}{w}(q^2 - 1)^w/(q^n + 1)$. Dividing by the $q - 1$ scalar multiples and the $2n$ dihedral images gives $\lambda = E/(2n(q-1))$, a heuristic expected number of "bad orbits" in a symmetric circulant: the division treats the orbits as independent, and symmetric circulants are not random Lagrangians. Table II lists λ for the scanned cells. Hits occur at $\lambda = 3.5$ and 11.4, and no AME state appears in the exhaustive scans with $\lambda \ge 20$. The Poisson guess $e^{-\lambda}$ is a rough guide, and its calibration depends on the normalization. With the dihedral factor $2n$ it overestimates the $n = 15$ hit rate by a factor of about 30 and predicts about 27 hits for AME$(13, 4)$, where the scan finds none. With a factor n instead, it predicts about 290 hits for $n = 15$ (336 found) and fewer than one for AME$(13, 4)$. Even n is much harder than the neighbouring odd n: λ is 51 for $(16, 8)$ against 3.5 and 11 for $(15, 8)$ and $(17, 8)$. Nor does an odd-n state project down to the even case (Sec. II). Known cells such as AME$(14, 7)$ and AME$(18, 7)$ are realized by Paley-type (quadratic-residue, bordered) circulants, where algebraic structure beats the first-moment count.

TABLE II. Exhaustive scans of symmetric circulant graph states over $\mathbb{F}_q$ (first coefficient normalized). $E/(q-1)$ is the expected number of nonzero stabilizers of party weight at most $\lfloor n/2 \rfloor$, up to scalars, in a uniformly random $\mathbb{F}_q$-linear stabilizer state (exact, denominator $q^n + 1$; Sec. III), and $\lambda = E/(2n(q-1))$ is the heuristic orbit count of Sec. III. "Circulants" is the family size $(q^{\lfloor n/2 \rfloor} - 1)/(q - 1)$, every member of which was tested.

(n, q)	$E/(q-1)$	λ	circulants	AME found
(15,8)	104	3.5	299,593	336 hits (28 orbits)
(17,8)	388	11.4	2,396,745	48 hits (4 orbits)
(16,8)	1,644	51.4	2,396,745	0
(19,8)	1,450	38.2	19,173,961	0
(16,7)	1,853	57.9	960,800	0
(19,7)	1,859	48.9	6,725,601	0
(16,5)	2,410	75.3	97,656	0
(19,5)	3,311	87.1	488,281	0
(19,9)	1,159	30.5	48,427,561	0
(19,11)	785	20.7	235,794,769	0
(13,4)	102	3.9	1,365	0
(15,4)	360	12.0	5,461	0

For $(16, 8)$ the zero is in fact forced. Let q be even, $4 \mid n$, $H \subset \mathbb{Z}_n$ the even positions, and $c(x) = c_{\delta(0,x)}$. The block $\Gamma[H, \mathbb{Z}_n \setminus H]$ is the matrix of multiplication by $\mu = \sum_{u\in H} c(1 + u)\, u$ in the group algebra $\mathbb{F}_q[H]$, so it is invertible only if the augmentation $\varepsilon(\mu) = \sum_{x \text{ odd}} c(x)$ is nonzero. Since $n/2$ is even, no odd x satisfies $x = -x$, and the terms cancel in pairs, $c(x) + c(-x) = 0$. So no symmetric circulant graph state over $\mathbb{F}_{2^r}$ on $\mathbb{Z}_n$ with $4 \mid n$ is AME. The same argument applies to every regular group with an index-2 subgroup that contains all involutions (found by one agent for cyclic and abelian groups, re-derived and extended to non-abelian groups by the auditing agent).

IV. LOCAL DIMENSION 16

The Huber–Wyderka table stops at $d = 11$. For $12 \le d \le 64$ and $n \le 40$ our verifier uses a snapshot in which a cell counts as existing only if a published construction gives it: Reed–Solomon (MDS) codes for $n \le d+1$ with d a prime power, the seven-party states of Ref. [16], the constructions of Refs. [17, 18], Kitamura's states, a few further published statements (for example AME$(4, d)$ from orthogonal Latin squares), and closure under projection (Sec. II) and under tensor products on the same parties (which also yields four cells from our AME$(15, 8)$ and AME$(17, 8)$, such as AME$(15, 40)$). The

beyond-table part of the snapshot has 947 existing cells and 1,120 cells that none of these constructions gives; we say that such a cell is *not covered by the snapshot*. This is a statement about the archived snapshot (released with this paper, together with the program that builds it), not about the whole literature, so every state below needed its own literature check, and priority is a separate question from existence. For $d = 16$, Reed–Solomon codes give AME$(n, 16)$ up to $n = 17$, and the first cells not covered by the snapshot are $n \geq 18$.

Theorem 2. *The four stabilizer states of Table III are AME*$(18, 16)$*, AME*$(19, 16)$*, AME*$(20, 16)$ *and AME*$(21, 16)$*.*

Verification. The frozen verifier (version 2, Appendix A) checked every cut of the $4n$-qubit $\mathbb{F}_2$ stabilizer matrices in the candidate files. The auditing agent re-checked each of them with its own $\mathbb{F}_2$ elimination and rebuilt each object at the field level with its own arithmetic: every $\Gamma[A, A^c]$ has full rank over $\mathbb{F}_{16}$ for the two graph states, and for the two codes $A\bar{A}^T = I$ (characteristic 2) and all $\binom{n}{n/2} - 1$ square minors of A are nonzero. Changing one coefficient of AME$(18, 16)$ destroys the Hermitian property and leaves 154 singular minors. The script accompanying this paper confirms that the data of Table III describe the certified candidate files. □

How they were found. AME$(19, 16)$ came from random sampling of symmetric circulants over $\mathbb{F}_{16}$ (five hits in 6×10^6 samples). For $n = 21$ the symmetric circulants invariant under groups of multipliers (families with 273, 4369 and 1.8×10^7 members) gave nothing. The solution has a twisted symmetry instead: the multiplier $i \mapsto 8i$ combined with the Frobenius map $z \mapsto z^4$, i.e. $c_{8j} = c_j^4$. The family with this symmetry has 1 048 576 members and 36 hits, and every hit lies in a single orbit under multipliers, Frobenius maps and scaling. AME$(18, 16)$ is a circulant construction of the kind introduced by Ball [13], who found examples for $q \leq 7$ and searched only symmetric circulants for 18 parties; here A is a non-symmetric 9×9 circulant. We parametrized the unitary circulants by their eigenvalues over $\mathbb{F}_{2^{24}}$, and 0.44% of them are MDS (45 of the first 10 200). AME$(20, 16)$ uses a symmetric unitary A made of 5×5 circulant blocks that is invariant under a fixed-point-free semilinear involution (exchange of the two block rows and columns, the reflection $x \mapsto -x$ of $\mathbb{Z}_5$, and conjugation of the entries). The family has about 4.3×10^9 members and a hit rate of about 1.7×10^{-6}. Projecting one party of AME$(20, 16)$ gives another AME$(19, 16)$.

Why no generalized Reed–Solomon code gives them. An $\mathbb{F}_{256}$-linear construction of AME$(n, 16)$ needs a Hermitian self-orthogonal MDS code $[n, \lfloor n/2 \rfloor]_{256}$ (self-dual for even n, a quantum MDS code $[[n, 1, (n+1)/2]]_{16}$ for odd n). For q even and $q/2 \leq k \leq q - 1$, Ball and Vilar [38] show that a Hermitian self-orthogonal truncated generalized Reed–Solomon (GRS) code $[n, k]_{q^2}$ needs $n \geq q(k+1-q/2)$; for $q = 16$ this is $n \geq 32$ for $k = 9$ and $n \geq 48$ for $k = 10$. Hence no GRS code of length n gives these cells directly. One indirect route remains for AME$(18, 16)$: a pure quantum MDS code $[[17, 1, 9]]_{16}$, i.e. a Hermitian self-orthogonal MDS code $[17, 8]_{256}$, extends to an AME$(18, 16)$ state, and for $k = 8 = q/2$ the bound only requires $n \geq 16$. Whether a GRS code of this kind exists is undecided (searches with evaluation points in the 17th roots of unity and in $\mathbb{F}_{16} \cup \{\infty\}$ found none). Our code is not of this kind: its shortening, a Hermitian self-orthogonal MDS code $[17, 8, 10]_{256}$, is not GRS, because the columns of its generator matrix lie on 19 linearly independent quadrics, whereas the columns of a GRS code of length $2k+1 = 17$ lie on a normal rational curve and so on 21.

Novelty and what remains uncovered. We found no earlier construction of these four cells in the literature we could reach (full texts or abstracts; tables of quantum MDS codes for $q = 16$ were not reachable for us). We therefore regard them as new, with two caveats: absence from the sources we read is not a proof of absence, and the revision announced in Ref. [20] (Table I) would cover them, and every AME$(N, 16)$ with $N \leq 32$, if it includes $q = 16$. AME$(22, 16)$, AME$(23, 16)$ and AME$(24, 16)$ remain uncovered, by the snapshot and by our searches. Searches over circulants with multiplier and Frobenius symmetry and over extensions of the AME$(21, 16)$ graph by one party with a shift-invariant row found nothing, and the same Ball–Vilar bound ($n \geq 64$ for $k = 11$, $n \geq 80$ for $k = 12$) excludes GRS codes there. The shadow inequality in purity form [7] excludes none of the cells not covered by the snapshot.

V. COUPLING: COMPOSITE LOCAL DIMENSIONS AT ODD n

A state of N parties is *k-uniform* if every k-party marginal is maximally mixed; it is AME if it is $\lfloor N/2 \rfloor$-uniform.

Lemma 3 (coupling)**.** *Let $k \geq 1$, let $|G_a\rangle$ be a k-uniform state of $2k + 2$ parties of local dimension a, and let $|G_b\rangle$ be an AME$(2k + 2, b)$ state with $a \leq b$. Write L for the last party and $|A_t\rangle = \langle t|_L |G_a\rangle$, $|B_t\rangle = \langle t|_L |G_b\rangle$ for $t < a$. Then*

$$|\psi\rangle = \sqrt{b} \sum_{t=0}^{a-1} |A_t\rangle \otimes |B_t\rangle, \tag{2}$$

in which party $i \leq 2k + 1$, of dimension ab, holds party i of both states, is an AME$(2k + 1, ab)$ state.

Proof. Fix a set S of k of the $2k + 1$ parties and let T be the other $k + 1$. Then $\rho_S = b \sum_{t,t'} \mathrm{Tr}_T\, |A_t\rangle\langle A_{t'}| \otimes \mathrm{Tr}_T\, |B_t\rangle\langle B_{t'}|$. Since $\mathrm{Tr}_T\, |B_t\rangle\langle B_{t'}| = \langle t|_L\, \rho_{S\cup L}^{G_b} |t'\rangle_L$ and $\rho_{S\cup L}^{G_b} = \mathbb{1}/b^{k+1}$ (G_b is AME and $|S \cup L| = k + 1$), this factor is $\delta_{tt'}\mathbb{1}/b^{k+1}$. Hence $\rho_S =$

TABLE III. The four states with $d = 16$. Field elements are integers whose binary digits are the coordinates in the polynomial basis, $\mathbb{F}_{16} = \mathbb{F}_2[x]/(x^4 + x + 1)$ and $\mathbb{F}_{256} = \mathbb{F}_2[x]/(x^8 + x^4 + x^3 + x^2 + 1)$; in $\mathbb{F}_{256}$ the conjugation is $z \mapsto z^{16}$. Graph states: $\Gamma_{ij} = c_{\delta(i,j)}$ as in Sec. III. Hermitian codes: generator $[I \,|\, A]$ over $\mathbb{F}_{256}$ (Sec. II), A built from circulants with the given first rows. "Cuts" is the number of bipartitions into $\lfloor n/2 \rfloor$ and $\lceil n/2 \rceil$ parties, all of which the verifier checked. "Found by" names the agent or, for AME(19, 16), the coordinating session (Sec. VII).

cell	object	data	cuts	found by
AME(18, 16)	Hermitian self-dual MDS $[18, 9, 10]_{256}$; A a 9×9 circulant	first row of A: $(92, 74, 70, 173, 117, 59, 208, 212, 182)$	24,310	extend
AME(19, 16)	graph state, symmetric circulant on $\mathbb{Z}_{19}$ over $\mathbb{F}_{16}$	$c = (1, 5, 11, 7, 6, 12, 4, 0, 4)$	92,378	coordinator
AME(20, 16)	Hermitian self-dual MDS $[20, 10, 11]_{256}$; $A = \left(\begin{smallmatrix} C_{00} & C_{01} \\ C_{01}^T & C_{11} \end{smallmatrix}\right)$, 5×5 circulants	first rows of C_{00}: $(78, 231, 216, 216, 231)$; of C_{01}: $(215, 146, 221, 152, 79)$; of C_{11}: $(78, 117, 65, 65, 117)$	92,378	extend
AME(21, 16)	graph state, symmetric circulant on $\mathbb{Z}_{21}$ over $\mathbb{F}_{16}$, $c_{8j} = c_j^4$	$c = (0, 12, 6, 2, 10, 1, 1, 0, 1, 3)$	352,716	extend

$\left(\sum_{t<a} \mathrm{Tr}_T \,|A_t\rangle\langle A_t|\right) \otimes \mathbb{1}/b^k = \rho_S^{G_a} \otimes \mathbb{1}/b^k = \mathbb{1}/(ab)^k$, because $\sum_{t<a} \mathrm{Tr}_T \,|A_t\rangle\langle A_t| = \mathrm{Tr}_{T\cup L} \,|G_a\rangle\langle G_a|$ and G_a is k-uniform. The condition $a \le b$ is needed so that the labels $t < a$, which run over a basis of party L of G_a, are also basis labels of party L of G_b. □

The proof uses no structure of G_a and G_b, so the lemma holds for arbitrary states, and tensor products of k-uniform (or AME) states on the same parties may serve as ingredients of composite dimension. For stabilizer ingredients with m digits of dimension p per party, G_a is k-uniform if and only if every set of $N - k$ of its $N = 2k + 2$ parties has a full-rank column block of the $mN \times 2mN$ generator matrix (Lemma 1 with $N - k$ in place of $\lceil N/2 \rceil$). Version 2 of our verifier has a coupling type that runs this test on G_a, the AME test on G_b, and concludes AME$(2k + 1, ab)$ by Lemma 3.

Instances and related constructions. Kitamura's AME(11, 10) [21] is the case $k = 5$, $a = 2$, $b = 5$: G_a is the 12-qubit dodecacode state ($[[12, 0, 6]]_2$, 5-uniform [28]) and G_b the AME(12, 5) state of Ref. [17]. The seven-party states AME$(7, 2m)$, m odd, of Ref. [16] have a similar two-term form with a different proof. Related gluing constructions keep a single local dimension in both factors (Raissi *et al.* [30]) or produce parties of different dimensions (Shen and Chen [29]; Ball and Zhang [31]). We did not find the general statement of Lemma 3 in the literature.

Ingredients. The 5-uniform states of 12 qubits and of 12 qutrits come from the dodecacode and from the extended ternary Golay code $[12, 6, 6]_3$ [34]. For $a = 4$ we give explicit $\mathbb{F}_{16}$-linear Hermitian self-dual codes (stabilizer states of ququarts); the code tables [23] give only bounds. The 6-uniform $[[14, 0, 7]]_4$ state has generator $[I_7 \,|\, A]$ with A the 7×7 circulant over $\mathbb{F}_{16}$ (labels as in Table III) with first row $(0, 1, 2, 15, 6, 2, 7)$; 840 of the 20 475 unitary circulants of this size give a 6-uniform state. The 7-uniform $[[16, 0, 8]]_4$ state has generator $[I_8 \,|\, A]$ with A a bordered $\mathbb{Z}_7$ circulant over $\mathbb{F}_{16}$: $A_{00} = 2$, $A_{0j} = A_{j0} = 4$ for $1 \le j \le 7$, and the remaining 7×7 block is the circulant with first row $(2, 1, 1, 0, 2, 8, 5)$. For G_b, Reed–Solomon codes cover $2k + 2 \le b + 1$. The other cases are Hermitian self-dual MDS codes: AME(12, 5) from Ref. [17], AME(12, 7) a $\mathbb{Z}_6$ circulant, AME(12, 8) a $\mathbb{Z}_3$-block circulant, AME(12, 9) a bordered $\mathbb{Z}_5$ circulant, AME$(14, b)$ for $b = 5, 7, 11$ bordered $\mathbb{Z}_6$ circulants, AME(14, 9) a submatrix of the kernel of Ref. [18], AME(16, 9) the parent matrix of that reference, and AME(16, 11), AME(16, 13) bordered $\mathbb{Z}_7$ circulants. AME(16, 11) was known to exist [15], but we know no published construction; ours diagonalizes the circulant over $\mathbb{F}_{11^6}$, where $x^7 - 1$ splits, and scans the resulting 1.6×10^7 unitary members exhaustively (84 are MDS). The generator matrices of all 27 distinct ingredients (in the canonical $\mathbb{F}_p$ form that the verifier checks and at field level, with field conventions), the map from cells to ingredients, the exact rank-check counts for every ingredient, and negative controls are in the release (`coupling/`); a standalone checker that shares no code with the verifier re-checks all 28 certificates in a few seconds.

Novelty. None of the 26 cells follows from known cells by a tensor product on the same parties: every factorization $d = d_1 d_2$ involves a factor AME(n, d_i) that is not

TABLE IV. The 26 certified coupling cells AME(n, d), $n = 2k+1$, $d = ab$, grouped by the k-uniform ingredient G_a of $2k+2$ parties and dimension a. Without a mark, G_b = AME$(2k+2, b)$ is a Reed–Solomon state ($2k + 2 \le b + 1$); † marks an explicit Hermitian self-dual MDS code (see text). Calibration runs also rebuilt AME(5, 10), AME(11, 10).

n	k	G_a (a)	$d = ab$
11	5	dodecacode $[[12, 0, 6]]_2$ (2)	$14^\dagger$, $18^\dagger$, 22, 26, 34, 38, 46, 54, 58, 62
11	5	ternary Golay $[[12, 0, 6]]_3$ (3)	$15^\dagger$, $21^\dagger$, $24^\dagger$, 33, 39, 48, 51, 57
13	6	$[[14, 0, 7]]_4$, circulant (4)	$20^\dagger$, $28^\dagger$, $36^\dagger$, $44^\dagger$, 52
15	7	$[[16, 0, 8]]_4$, bordered $\mathbb{Z}_7$ (4)	$36^\dagger$, $44^\dagger$, $52^\dagger$

covered by the snapshot or does not exist. Two further cells follow from Lemma 3 and tensor products but are not certified, because version 2 of the verifier accepts only prime-power stabilizer ingredients: AME$(11, 42)$, from $a = 6$ (the tensor product of the dodecacode and Golay states) and $b = 7$, and AME$(11, 60)$, the tensor product of AME$(11, 15)$ and AME$(11, 4)$. We do not list them as results. The remaining odd cells with a factorization $d = ab$ and a known AME$(2k + 2, b)$ need k-uniform ingredients that nobody has built, for example a 6-uniform $[[14, 0, 7]]_3$ (eight cells at $n = 13$), a $[[16, 0, 8]]_3$ and an $[[18, 0, 9]]_4$. A heuristic first-moment count over random $\mathbb{F}_p$-linear graph states (cuts treated as independent) gives expected numbers e^{-69}, e^{-512} and $e^{-2.3\times10^3}$ for these three, so unstructured search over graph states is hopeless. Over $\mathbb{F}_{q^2}$-linear Hermitian codes the naive count is favourable for $[[14, 0, 7]]_3$, but a randomized depth-first search through about 1.2×10^5 partial $\mathbb{F}_9$-linear codes found none (every completion examined had exactly 16 words of weight 6), and the structured families tried so far found none either.

VI. STABILIZER AME(8,4)

A. Which stabilizer formalisms remain

$\mathbb{F}_4$-linear stabilizer states (Pauli operators $X(a)Z(b)$, $a, b \in \mathbb{F}_4^8$) cannot be AME$(8, 4)$: the code tables give $d \le 4$ for $[[8, 0, d]]_4$ [23], and this is also a special case of Theorem 3 below. A ququart, however, admits other error bases. Let each party carry a group-type nice error basis $\{E_g\}_{g\in G}$, $|G| = 16$, with $E_gE_h = \omega(g, h)E_{gh}$ [32, 33]. A stabilizer state is the unique joint eigenvector of $\{E_g : g \in K\}$ for a subgroup $K \le G^8$ of order 4^8. As above, ρ_A is maximally mixed if and only if no $g \ne e$ in K is supported in A.

Proposition 1. *If such a state is AME$(8, 4)$, then $G \cong \mathbb{Z}_2^4$, and K is a Lagrangian subspace S of $V = \bigoplus_{j=1}^{8} \mathbb{F}_2^4$ for the sum of the local symplectic forms (a qubit-pair stabilizer state) in which every nonzero vector has party weight at least 5.*

Proof. For any set T of four parties, the projection $\pi_T : K \to \prod_{j\in T} G_j$ has as kernel the elements supported on the other four parties, so it is injective. Since $|K| = 16^4$, it is an isomorphism. Identify $K = G_1 \times \cdots \times G_4$ through $\pi_{\{1,2,3,4\}}$. The coordinate of party 5 is then a homomorphism $\prod_k \alpha_k(g_k)$, where the $\alpha_k : G_k \to G_5$ have pairwise commuting images. Each α_k is injective, by bijectivity of $\pi_{\{1,2,3,4,5\}\setminus\{k\}}$, hence surjective. So $G_5 = \alpha_1(G_1)$ commutes with $\alpha_2(G_2) = G_5$, and G_5 is abelian. By symmetry every G_j is abelian, and cancellation in $G_u \times G_T \cong G_v \times G_T$ makes them isomorphic. An abelian index group of a nice error basis is of symmetric type $A \times A$ with a nondegenerate alternating bicharacter [33]. With $|G| = 16$, this means $\mathbb{Z}_2^4$ or $\mathbb{Z}_4^2$. For $\mathbb{Z}_4^2$ (Weyl–Heisenberg over $\mathbb{Z}_4$), $K \cong \mathbb{Z}_4^8$ is free, because π_T is a bijection onto $\mathbb{Z}_4^8$. Its reduction $\bar{K} = K \bmod 2 \subset (\mathbb{F}_2^2)^8$ then has dimension 8, it is isotropic because commutation mod 4 implies commutation mod 2, and $2s \in K$ has the party support of $\bar{s}$. Thus $\bar{K}$ would be a stabilizer AME$(8, 2)$ state, which does not exist [6, 8]. (Reference [22] makes the same remark for $\mathbb{Z}_4$-codes with these parameters.) □

A related Singleton-type argument shows more generally that a $\mathbb{Z}_{p^2}$ stabilizer AME(n, p^2) requires a stabilizer AME(n, p). From now on, $S \subset V$ is a qubit-pair Lagrangian with minimum party weight 5, and $\pi_T(S) = \mathbb{F}_2^{16}$ for every set T of four parties. Taking $T = \{1, 2, 3, 4\}$ gives the normal form $S = \{(u, Mu)\}$ with $M \in \mathrm{Sp}(16, 2)$, in which all 69 party-aligned square blocks of M (including M itself) are invertible. These conditions encode the 35 balanced cuts.

The scope of Proposition 1 is the stabilizer model defined above: a group-type nice error basis on each party and a stabilizer group of order 4^8. It says nothing about non-stabilizer AME$(8, 4)$ states, nor about states defined through unitary error bases that are not of group type; the nonexistence results below inherit this assumption.

B. No qubit-pair stabilizer AME(8,4): an independent confirmation

Theorem 3 (Ball, Moreno, and Simoens [22])**.** *There is no qubit-pair stabilizer AME$(8, 4)$ state: every Lagrangian $S \subset V$ contains a nonzero vector of party weight at most 4. Equivalently, there is no $[[8, 0, 5]]_4$ stabilizer code.*

Reference [22] works in exactly this model: Pauli operators labelled by $\mathbb{F}_{2^h}$ and additive ($\mathbb{F}_2$-linear) stabilizers, here with $h = 2$. It proves that no $[[7, 1, 4]]_4$ code exists, by a computer classification of $[[6, 2, 3]]_4$ codes in the geometry of symplectic polar spaces, and derives Theorem 3 from that. With Proposition 1, Theorem 3 excludes every group-type stabilizer AME$(8, 4)$ state. We confirm Theorem 3 by an exhaustive search that is organized differently and was implemented twice. This is an independent check, not a new theorem. We describe the reduction and the two implementations so that the check can itself be checked.

Frames. Let $A = \{0, 1, 2, 3\}$ and $B = \{4, 5, 6, 7\}$, let $P = \mathbb{F}_2^4$ carry the alternating form J of one ququart, and let P^A, P^B be the sums over the parties of A and B with the sum form ω. By the normal form above, $S = \{(u, Mu) : u \in P^A\}$ with $M : P^A \to P^B$ a linear bijection. In characteristic 2, isotropy of S says that M is an *isometry* for ω. Write A_j for the copy of P of party $j \in A$, W_l for the copy of P of party $4 + l$, and

$$V_j := M(A_j) \subset P^B, \qquad V_J = \bigoplus_{j\in J} V_j, \qquad W_L = \bigoplus_{l\in L} W_l.$$

Lemma 4. *(a) $V_0, \ldots, V_3$ are mutually ω-orthogonal nondegenerate 4-dimensional subspaces with $P^B = V_0 \oplus \cdots \oplus V_3$. Conversely, every such decomposition comes from a Lagrangian S, which is determined up to the local Cliffords $\mathrm{Sp}(4,2)$ of the parties of A. (b) For S as in (a), S has minimum party weight ≥ 5 if and only if $V_J \cap W_L = 0$ for all $J, L \subseteq \{0,1,2,3\}$ with $|J| + |L| = 4$.*

Proof sketch. (a) M is an isometry and a bijection, and the A_j are orthogonal and nondegenerate. Conversely, any isometries $E_j : (P, J) \to (V_j, \omega|_{V_j})$ define M, and two choices differ by $\mathrm{Sp}(4,2)$ on party j. (b) For $0 \neq v \in P^B$ the stabilizer element $(M^{-1}v, v)$ has A-support $\{j : v_j \neq 0\}$, where $v = \sum_j v_j$ with $v_j \in V_j$, and B-support $\{l : \pi_l v \neq 0\}$. So $v \in V_J \cap W_L$ if and only if the A-support of $(M^{-1}v, v)$ is contained in J and its B-support is contained in L. A nonzero element of party weight ≤ 4 has A-support J_0 and B-support L_0 with $|J_0| + |L_0| \leq 4$, which can be enlarged to $J \supseteq J_0$, $L \supseteq L_0$ with $|J| + |L| = 4$. Conversely, a nonzero $v \in V_J \cap W_L$ gives an element of party weight $\leq |J| + |L| = 4$. □

Of the $16+36+16 = 68$ conditions of (b) with $1 \leq |J| \leq 3$ (the others are trivial), the cases $|J| = 1$ say that each $\pi_l|_{V_j} : V_j \to W_l$ is a bijection. So every V_j is a graph over W_0, $V_j = \{(w, \theta_{j1}w, \theta_{j2}w, \theta_{j3}w)\}$ with $\theta_{jl} \in \mathrm{GL}(4,2)$, and the orthogonality $V_i \perp V_j$ reads $\sum_{l=0}^{3} \theta_{il}^T J \theta_{jl} = 0$ ($\theta_{i0} = 1$), which is *linear* in $\theta_j = (\theta_{j1}, \theta_{j2}, \theta_{j3}) \in \mathbb{F}_2^{48}$ for fixed V_i.

Gauge. The group $\mathrm{Sp}(4,2) \wr S_4$ (local Cliffords and permutations of B), together with S_4 renumbering the V_j, preserves ω, orthogonality and the family of conditions. It therefore suffices to take V_0 from a set of representatives and let V_1, V_2, V_3 range over everything. A V_0 satisfying the $|J| = 1$ conditions carries four nondegenerate alternating forms $Q_l = (\pi_l|_{V_0})^* J$ (28 choices each) whose sum $\omega|_{V_0}$ is nondegenerate, and the gauge orbit of V_0 is the orbit of $(Q_0, \ldots, Q_3)$ under $\mathrm{GL}(V_0) \times S_4$. Of the 614,656 tuples, 268,800 have a nondegenerate sum, and they fall into 10 orbits (Table V). For $n = 4$ and $n = 6$ parties the same construction gives one and five orbits.

Enumeration. For each of the ten representatives V_0 we solve the linear orthogonality system for θ_1 (a 32-dimensional affine space for every V_0), and test each of its 2^{32} points in Gray-code order for $\theta_{1l} \in \mathrm{GL}(4,2)$, for nondegeneracy, and for every condition of Lemma 4(b) that involves only V_0, V_1 (rank tests). The surviving V_1 (24,128 to 28,960 per class) are extended in the same way to V_2 (a 16-dimensional affine space of candidates for each V_1), and $V_3 = (V_0 + V_1 + V_2)^\perp$ is then forced. No pruning is heuristic: each test is a necessary condition of Lemma 4 applied to a prefix of the frame.

Table V gives the result: 267,980 valid V_1 in total, and *no* valid V_2 for any of them (about 25 s of C per class). Hence there is no frame, which confirms Theorem 3. The confirmation is computer-assisted and rests on the enumeration alone. A naive independence count predicts fewer than one survivor in total, but the intermediate counts do not follow the independence model, so we do not use it as evidence. Without the four conditions with $J = \{0,1,2\}$, 24,234 triples (V_0, V_1, V_2) survive, and the independence model would predict about 220 of them to satisfy those four conditions as well; none does. In every one of these triples, V_3 contains between 3 and 15 nonzero vectors that meet at most three of the four parties of B. We have no explanation of this pattern.

TABLE V. Frame search for a qubit-pair stabilizer AME(8, 4). For each of the ten gauge classes of V_0: the size of its orbit of form tuples $(Q_0, \ldots, Q_3)$ (total 268,800), and the number $\#V_1$ of V_1 such that (V_0, V_1) satisfies every condition of Lemma 4(b) that involves only these two subspaces (total 267,980). In every class the number of valid (V_0, V_1, V_2) is 0, where valid means that V_0, V_1, V_2 satisfy every condition of Lemma 4(b) that involves only V_0, V_1, V_2 (including the four conditions with $J = \{0,1,2\}$, which are those on V_3). Both implementations give these numbers.

class	0	1	2	3	4
orbit	80,640	60,480	40,320	20,160	20,160
$\#V_1$	26,556	24,128	24,128	28,960	24,488
class	5	6	7	8	9
orbit	20,160	13,440	10,080	2,016	1,344
$\#V_1$	28,960	24,488	28,960	28,656	28,656

Checks. The same code run at other parameters reproduces known facts. For $n = 4$ it counts $8640 \cdot 720^2 = 4{,}478{,}976{,}000$ qubit-pair stabilizer AME(4, 4) states, which equals the order of $\mathrm{Sp}(4,2) \wr S_4$ divided by the 1440 automorphisms of a representative (Appendix B). For $n = 6$ all five classes of V_0 complete to AME(6, 4) states (party weight enumerator $[1, 0, 0, 0, 225, 1080, 2790]$). The predicted total, $54400 \cdot 720^5$, is a fraction 1.461×10^{-5} of all Lagrangians, and 4×10^7 uniformly random Lagrangians gave 574 AME states against 584 ± 24 expected. At $n = 8$, with the six conditions for $J = \{0,2\}$, $|L| = 2$, and their duals removed, the search produces solutions at once: genuine stabilizer states with minimum party weight 4 and as few as one bad cut out of 35. A SAT encoding of the normal form finds a solution of the relaxed problem and one for each of the five classes at $n = 6$, but it cannot decide $n = 8$ (it also times out on the satisfiable AME(10, 4) control), so there is no UNSAT certificate. The ten representatives of V_0 (for both implementations), the completeness argument for the reduction, both enumeration codes, the expected per-class counts with SHA-256 hashes, and the positive controls are in the release (`frames/`), with a script that replays them from a fresh copy.

Second implementation. An auditing agent wrote a second C implementation with a different enumeration order (depth-first over lexicographically least bases of V_1, V_2 inside weight-filtered candidate sets) and its own classification of V_0, by $\mathrm{Sp}(4,2)$-orbits. It finds the same ten classes (its orbit sizes, computed with the first form fixed, are those of Table V divided by 28), the same $\#V_1$ in every class, the same 24,234 triples (class by class), and no frame. Its subspace enumerator reproduces the Gaus-

sian binomial counts exactly, and its result is invariant under random gauge transformations. The two searches share the reduction of Lemma 4, which the auditor rederived in writing but did not discover independently. The method-independent confirmation is Ref. [22].

C. What remains

What remains open for AME$(8,4)$ are non-stabilizer states, including states built from unitary error bases that are not of group type (for example shift-and-multiply bases from Hadamard matrices and Latin squares). We do not know a structured ansatz for these with a searchable parameter space. Random-start continuous solvers fail even where AME states are known to exist (AME$(6,4)$, AME$(4,6)$). The ratio of their plateau to the random baseline is 0.345 at $(8,3)$, where no AME state exists, and 0.350 at $(8,5)$, where one does, so a plateau at $(8,4)$ carries no information.

Other cells. Hermitian self-dual GRS codes do not exist for $n > q+1$ (see Ref. [37] and references therein). An $\mathbb{F}_{64}$-linear stabilizer AME$(16,8)$ is a Hermitian self-dual MDS $[16,8,9]_{64}$ code, equivalently (by shortening) a Hermitian self-orthogonal MDS $[15,7,9]_{64}$ code. No truncated GRS $[15,7]_{64}$ code is Hermitian self-orthogonal: by the criterion of Ref. [38] and its Theorem 4.3, such codes need length at least $q(k+1-q/2) = 32$ for $q = 8$, $k = 7$, and our own enumeration in a trace code reproduces this bound. Searches over many further structured families, exhaustive unless stated otherwise in the project repository (not public; available on request), found no new states. These families are Hermitian codes from pure and bordered group circulants (cyclic, $\mathbb{Z}_3^2$, $\mathbb{Z}_2^3$, $\mathbb{Z}_2 \times \mathbb{Z}_4$, and modular group algebras in characteristic 2; compare Ref. [13]), constacyclic and twisted codes, elliptic codes, and graph codes with abelian Cayley, multiplier, orbital and twisted symmetry. The graph-code families covered the cells not covered by the snapshot with $13 \le n \le 20$ and $d \in \{3,4,5,7,8,9,11\}$; the Hermitian families ran on selected even-n cells up to $n = 24$. The second phase of the run added graph codes invariant under large transitive groups (projective linear and Mathieu groups among them), Hermitian codes from non-abelian regular groups, codes on unions of orbits, $\mathbb{F}_2$-linear block families with two or three qubits per party (some of them sampled rather than exhausted), and a complete scan of the pure $\mathbb{Z}_{10}$ group circulants over $\mathbb{F}_{64}$ for AME$(20,8)$; none gave a new state. The repository lists all families with sizes.

VII. HOW THE RESULTS WERE OBTAINED

The search was run in three phases by swarms of language-model agents (Claude Sonnet 5.5) on one laptop, coordinated through a message board. Phase 1 (five agents, four hours) found AME$(15,8)$ and AME$(17,8)$ and carried out the symmetry analysis of Appendix B. Phase 2 (four agents, four hours) carried out the frame search of Sec. VI B, its second implementation, the parity argument of Sec. III and the further closures of Sec. VI. Phase 3 (three agents, four hours) worked beyond the table: one agent (extend) found AME$(18,16)$, AME$(20,16)$ and AME$(21,16)$, one (couple) the coupling cells of Sec. V, and the auditing agent checked them; AME$(19,16)$ was found by the coordinating session while it set up the phase. Permission rules prevented the agents from modifying the verifier, the record data and the configuration, and the protocol forbade editing each other's files. Only verdicts of the frozen verifier counted as results. The verifier is pinned by hash, and its tests include known AME states, every verdict path and brute-force cross-checks. In every phase an auditing agent re-checked the new states, the frame search and selected closures with independent code (the audit record lists which), wrote out or reviewed most of the proofs, and searched the literature; in phase 2 it wrote the second frame search and found Ref. [22], after both searches had finished. All 32 certified candidates (2 from phase 1, 30 from phase 3) have a frozen candidate file and an independent re-check (for the coupling cells, of both ingredients). Two red-team rounds against version 1 of the verifier (88 adversarial inputs in the second) and 27 red-team cases for the coupling type of version 2 (adversarial inputs and valid controls) produced no wrongly accepted candidate. The two states of phase 1 were verified 3.3 and 4.1 minutes after its start, and the two frame searches reported their null results 12 and 18 minutes after the start of phase 2. The scripts, the agents' notebooks and the certificates are in the project repository (not public; available on request). The release (see Data availability) contains what is needed to replay the certified states, the coupling certificates, every exhaustive search behind a count or a nonexistence statement in Secs. III–VI B and Appendix B, and the quadric and shadow tests. The hit rates of sampled discovery searches and the negative closures of Sec. VI other than the frame search are documented in the repository only. This manuscript was drafted by a language model (Claude Opus 5.5) from those records and checked against them.

VIII. CONCLUSION

AME$(15,8)$ and AME$(17,8)$ exist, as circulant graph states over $\mathbb{F}_8$. AME$(16,8)$ is now the gap between two existing cells, and no symmetric circulant over $\mathbb{F}_8$ can fill it. For $d = 16$ the frontier moves from $n = 17$ to $n = 21$, by circulant graph states over $\mathbb{F}_{16}$ and non-GRS Hermitian self-dual MDS codes over $\mathbb{F}_{256}$; AME$(22,16)$ is the next cell not covered by the snapshot. The coupling rule turns k-uniform states of small dimension and AME states of larger dimension into AME states of composite dimension at odd n, and its reach is limited by the k-uniform ingredients, such as a 6-uniform state of 14 qutrits, that are still missing. For AME$(8,4)$, no

stabilizer state exists for a group-type nice error basis: Proposition 1 reduces the question to qubit pairs, where Ball, Moreno and Simoens proved nonexistence [22] and our differently organized exhaustive search confirms it. AME$(8,4)$ itself remains open: non-stabilizer states, and error bases outside the group-type model, are not covered by Proposition 1 or by the searches. A natural next step is structured non-stabilizer constructions in the spirit of Refs. [14, 19].

DATA AVAILABILITY

The release accompanying this paper is at https://github.com/maxxfan/aixmathxphysics/tree/ame-states-v4/papers/ame-states/artifacts. It contains, with a `README`, `SHA256SUMS` and a replay script that runs from a fresh copy with Python 3 and a C compiler (24 of the 60 checks of `appendixB/` also need numpy; everything else uses the standard library only): the six explicit states with their printed field-level data and a field-level checker (`states/`); the 28 coupling certificates (the 26 new cells and two calibrations), their 27 distinct ingredients in canonical form and at field level, the ingredient-to-cell map, negative controls and a standalone checker (`coupling/`, `checker/`); both frame-search implementations with the ten gauge representatives, counts, hashes and controls (`frames/`); executable evidence for the sectors of Appendix B (`appendixB/`); the circulant scans of Table II, the quadric test of Sec. IV and the shadow test (`ancillary/`); and the three record snapshots used by the verifier, with their sources (`snapshot/`).

Appendix A: The verifier

Candidates are either stabilizer generator matrices over $\mathbb{F}_p$ (m qudits per party) or explicit state vectors with amplitudes in $\mathbb{Z}[\zeta_N]/\sqrt{D}$. For stabilizers, the verifier checks isotropy and rank. It then checks Lemma 1 for all $\binom{n}{\lceil n/2\rceil}$ sets in C, re-checks a seeded sample (all sets when small) in Python, and confirms for even n that every set and its complement agree (all sets when $\binom{n}{n/2} \le 2\times 10^5$, otherwise a sample). For states, the identities $d^a \sum_c a_{(r,c)}\overline{a_{(s,c)}} = D\,\delta_{rs}$ are evaluated at every primitive N-th root of unity modulo primes $P \equiv 1 \pmod N$. Since $\mathbb{Z}[\zeta_N]/(P) \cong \mathbb{F}_P^{\varphi(N)}$, vanishing at all roots means every coefficient is divisible by P. Taking primes whose product exceeds twice a rigorous coefficient bound then makes the identity exact. Verdicts compare the cell with a frozen snapshot of the AME table and the 2026 preprints. Version 2, used in phase 3, extends the snapshot beyond the table (Sec. IV) and adds a coupling type: two stabilizer candidates G_a, G_b of $2k+2$ parties with $a \le b$, of which G_a must pass the k-uniformity test and G_b the AME test of Sec. V; the verdict is then AME$(2k+1, ab)$ by Lemma 3. The coupling type was red-teamed with 27 cases (adversarial inputs and valid controls), with no wrong acceptance.

Appendix B: Symmetry sectors (cross-check)

Phase 1 of the run, before the frame search and before Ref. [22] was known to us, excluded every qubit-pair stabilizer AME$(8,4)$ state with a nontrivial symmetry. By Theorem 3 the two statements below hold trivially. We keep them because their proofs use counting arguments and searches that were written independently of the frame enumeration, so they cross-check its null result on the symmetric part of the search space. Let $\mathrm{Aut}(S)$ be the group of $\Omega = P_\sigma\, g$ ($\sigma \in S_8$ permutes parties, $g \in \mathrm{GL}(4,2)^8$) with $\Omega S = S$, and $\mathrm{Aut}_{\mathrm{loc}}(S)$ its subgroup with $\sigma = 1$. Because S projects onto every set of at most four parties, an automorphism acting trivially on S is the identity.

Proposition 2. *A qubit-pair stabilizer AME$(8,4)$ state has* $\mathrm{Aut}_{\mathrm{loc}}(S) = 1$.

Proof sketch. Since $|\mathrm{GL}(4,2)| = 2^6\cdot 3^2\cdot 5\cdot 7$, it suffices to exclude a local $\Omega \neq 1$ of prime order ℓ. For odd ℓ, $S = S^\Omega \oplus S_{\mathrm{nf}}$, where S^Ω lies in the fixed spaces $U_j = \ker(\omega_j - 1)$ and $S_{\mathrm{nf}} \neq 0$ lies in $W_j = \ker \Phi_\ell(\omega_j)$. Both summands have minimum party weight ≥ 5, so deleting any four parties is injective on each. This Singleton-type bound, applied to all assignments of per-party types $(\dim U_j, \dim W_j)$, leaves only $(2,2)^8$ and $(0,4)^8$ for $\ell = 3$, $(0,4)^8$ for $\ell = 5$, and $(1,3)^8$ for $\ell = 7$.

- $(2,2)^8$: S_{nf} is an $\mathbb{F}_4$-linear $[8,4,5]_4$ MDS code, which does not exist [34].
- $(1,3)^8$: S^Ω is a binary code of length 8, dimension ≥ 4 and distance ≥ 5, impossible since $A(8,5) = 4$.
- $(0,4)^8$: S is $\mathbb{F}_4$-linear for a fixed-point-free $\mathbb{F}_4$-structure ω_j on every party. Up to local $\mathrm{Sp}(4,2)$ there are three types: H (ω_j symplectic, 40 elements), S (ω_j self-adjoint, 12), and M (60). Mixed H/S types split S into a product across a cut, which is impossible. Type M makes S a module over a non-commutative chain ring of order 16. A Singleton argument on its socle again forces an $[8,4,5]_4$ MDS code, and a similar count excludes M parties mixed with H or S parties. Uniform H means S is a Hermitian self-dual $[16,8]_4$ code with paired coordinates. Uniform S is the $\mathbb{F}_4$-linear case of the code tables. Both uniform cases were excluded by exhaustive depth-first searches over systematic generator matrices $[I|M]$ over $\mathbb{F}_4$ with all party-aligned minors invertible ($MM^\dagger = I$ for type H). Each search was written twice, with different normalizations. The searches were calibrated on $n = 4$ and $n = 6$, where they reproduce independent brute-force counts (for example, 34 992 of the 77 760 unitary 4×4 matrices over $\mathbb{F}_4$), and against random

sampling at intermediate depths. The best H-type state has one bad cut out of 35.

For $\ell = 2$, $\omega_j - 1$ is nilpotent with rank $r_j \in \{0, 1, 2\}$, and the same counting leaves $(1)^8$ and $(2)^8$. In the first case $(\Omega - 1)S$ is a binary $[8, \geq 4, \geq 5]$ code, impossible. In the second, $(\Omega - 1)S$ is an orthogonal array $\mathrm{OA}(256, 8, 4, 4)$ of index one, which violates Bush's bound $k \leq t + 1$ for $t \geq s$ [35]. The counting parts of this proof were written by one agent (the auditor) and have not been re-derived independently; its exhaustive components (uniform H and S) were implemented twice. □

Proposition 3. *A qubit-pair stabilizer AME*$(8, 4)$ *state has* $\mathrm{Aut}(S) = 1$ *in* $\mathrm{GL}(4, 2) \wr S_8$.

Proof outline. (a) *Every automorphism is symplectic.* For $\Omega \in \mathrm{Aut}(S)$ the adjoint $\Omega^\dagger$ also preserves $S = S^\perp$. So $\Omega^\dagger\Omega = g^\dagger g$ is a local automorphism and equals 1 by Proposition 2, i.e. every $g_j \in \mathrm{Sp}(4, 2)$. (b) *Reduction to prime order.* If $\mathrm{Aut}(S) \neq 1$, it contains an element $\Omega = P_\sigma g$ of prime order $\ell \in \{2, 3, 5, 7\}$, and $\sigma \neq 1$ by Proposition 2. Since $\Omega^\ell = 1$, the product of the local maps around each cycle of σ is trivial. A local change of basis therefore makes Ω a pure permutation on the moved parties, with local maps $h_j \in \mathrm{Sp}(4, 2)$, $h_j^\ell = 1$, on the fixed parties. Because $\mathrm{Sp}(4, 2) \cong S_6$ has no element of order 7, $h = 1$ when $\ell = 7$. (c) *All sectors are empty.* Table VI lists every cycle type of σ and every class of twists h_j. Each sector is excluded by a counting argument or by an exhaustive computation. All exhaustive computations except the 7-cycle enumeration were implemented twice with independent code. The 7-cycle enumeration, the auditor's own work like Proposition 2, was validated against direct enumeration of the 2^{16} stabilizer elements. Every computational sector has a positive control. In most sectors it is a relaxed weight threshold of 4 that produces solutions; the searches for $(3^2, 1^2)$ with $h = 1$, for $(5, 1^3)$ and for $(7, 1)$ cover their whole space and exhibit states of minimum weight exactly 4. The control for the (2^4) sector was added when the release was prepared. The release (`appendixB/`) re-runs every computation of this appendix, makes the counting steps executable where possible, and lists for each row which steps remain hand derivations. □

In particular, no cyclic, dihedral, $\mathrm{AGL}(1, 8)$, $\mathrm{A\Gamma L}(1, 8)$, $\mathrm{PSL}(2, 7)$ or $\mathrm{AGL}(3, 2)$-symmetric ansatz yields a qubit-pair stabilizer AME$(8, 4)$, which Theorem 3 and the frame search now show without any symmetry assumption.

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TABLE VI. The prime-order sectors of $\mathrm{Aut}(S)$ for a qubit-pair stabilizer AME$(8,4)$, $S = S^\Omega \oplus S_{\mathrm{nf}}$ (odd ℓ) or $(\Omega-1)S \subseteq S^\Omega$ ($\ell = 2$). "Twist" is the local map h_j on the parties fixed by σ. Each counting row has a hand derivation, and each computational row has two independent implementations, except the $\ell = 7$ row (one implementation).

ℓ	σ (twist)	reduction and result
any	1	Proposition 2.
2	(2^4)	$S^\Omega = (\Omega-1)S = \{(w,w) : w \in C\}$ with C an AME$(4,4)$ qubit-pair code. All AME$(4,4)$ codes form a single orbit of $\mathrm{Sp}(4,2)^4 \rtimes S_4$: there are 4 478 976 000 of them, equal to the group order divided by the 1440 automorphisms of a representative. S is then determined by a symmetric form β on C; all 2^{36} forms give minimum weight ≤ 4.
2	$(2^3,1^2)$	$S^\Omega = (\Omega-1)S$ is determined by a pair of forms and an isometry: 8640 candidates, 10 up to gauge. A lift search for the rest of S finds 0 solutions (at threshold 4 a capped search finds more than 1.4×10^6).
2	$(2^2,1^4)$	S^Ω requires a nondegenerate subspace among the 432 subspaces of $(\mathbb{F}_2^2)^4$ with block weight ≥ 3; all 432 are degenerate.
2	$(2,1^6)$	Counting forces equal twist ranks $r \leq 2$. $r = 0$ and $r = 1$ fail by weight counting. $r = 2$ would need an OA$(16,6,4,2)$, which violates the Bose–Bush bound [36].
3	$(3,1^5)$, any	Counting over all 3^5 twist patterns: none survives.
3	$(3^2,1^2)$, $h = 1$	S_{nf} is the graph of one of the $\lvert\mathrm{GU}(4,2)\rvert = 77\,760$ $\mathbb{F}_4$-isometries; all have weight-4 words.
3	$(3^2,1^2)$, h with 2-dim fixed space	1152 completions T of S_{nf}, none admits a valid S^Ω.
3	$(3^2,1^2)$, mixed	Parity and weight counting.
3	$(3^2,1^2)$, h fixed-point-free	391 910 400 pairs (X_A, X_B) reduce to 7 classes and 138 completions T; none of the 720 $f \in \mathrm{Sp}(4,2)$ gives S^Ω (best: 3 bad cuts).
5	$(5,1^3)$, order-5 h	Counting forces two twisted fixed parties. S_{nf} is then one of the $333\,125 = (q+1)(q^3+1)(q^5+1)$ $(q=4)$ generators of the Hermitian polar space $H(5,16)$; the maximum minimum weight is 4.
7	$(7,1)$	$S_{\mathrm{nf}} = L \oplus L^\perp$ for an $\mathbb{F}_8$-subspace $L \subseteq \mathbb{F}_8^4$; all 5917 such L give minimum weight ≤ 4.

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